

PGO models in the envelope function and effective mass approximations

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Abstract. A recipe to design quantum devices that exhibit the theoretical pseudo-Gaussian oscillator electronic states properties is given. The algorithm is described *en detail* and is illustrated by the computation of a $\text{Mn}_x\text{Cd}_{1-x}\text{Te}$ ternary alloy pseudo-Gaussian heterostructure. The numerical procedure reaches beyond of pseudo-Gaussian models and can be used for designing epitaxial growth devices with desired electronic states structure. The calculations are carried out in the envelope function and effective mass approximations.

1 Introduction

Modern developments in epitaxial growth technology open the prospect of producing semiconductor layers systems with arbitrarily potential profiles. The knowledge of the *realistic* confinement potential profile is of great interest in designing quantum devices when a particular structure of electronic states or minibands is requested. Usually, the multiple quantum well system embedded in an electronic device is modeled either by a rectangular potential well [1] or a parabolic potential well [2]. The parabolic quantum well is suitable in a variety of applications [3–5] due to its distinctive properties: equally spaced electronic spectra and radiative transition or radiation absorption at the same oscillator frequency.

In reference [6] our group proposed a large family of models, called Pseudo-Gaussian Oscillators (PGO), whose potential have a Gaussian asymptotic behavior but approaching to that of the harmonic oscillator near $x = 0$. The PGO models provide almost equidistant bound states and resonances which have been calculated using the functional generating method. A three-dimensional isotropic model with similar PGO potentials [7] was reported as well as superlattice PGO models [8]. There are some distinctive features of PGO models. First, flexibility: as detailed in Section 2, by means of a three parameter fit, PGO potentials may replicate most natural or engineered semiconductor systems. Short-range interaction is a second advantage: PGO systems can be very narrow and with a finite number of equidistant levels. So, they can be used to model various defects in solids; 3D PGO isotropic potential is suitable for quantum dots description. Third, PGO models are always finite and analytical, thus avoiding singularities that may appear in other models (e.g.

when an infinite parabolic potential is cut off to simulate a real system).

All the above mentioned studies are focused on the theoretical and computational developments to reveal the new potential properties. However, they stopped before considering application to real semiconductor materials.

In this paper we propose to carry on the theoretical study of PGO models investigating methods to reproduce the features of the electronic systems of PGO using semiconductor materials. The calculations are based on the transfer matrix method (TMM) inside the envelope function and effective mass approximations [9].

The paper is organized as follows. We start in Section 2 with an overview of the PGO potential family and its virtues. In Section 3 we briefly describe the transfer matrix approach as a numerical tool for analyzing the system. Section 4 is intended to sketch a *recipe to tailor* a ternary semiconductor alloy heterostructure which exhibit the electronic states system of an arbitrarily PGO model.

2 Pseudo-Gaussian potentials

The one-dimensional Schrödinger equation for the harmonic oscillator (HO) of mass m and frequency ω can be written as:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2 x^2 \psi + V_0\psi = E\psi \quad (1)$$

where V_0 is an arbitrarily ground energy. A dimensionless variant of equation (1) may be obtained:

$$-\frac{d^2\psi}{d\xi^2} + \xi^2\psi + \lambda\psi = \nu\psi \quad (2)$$

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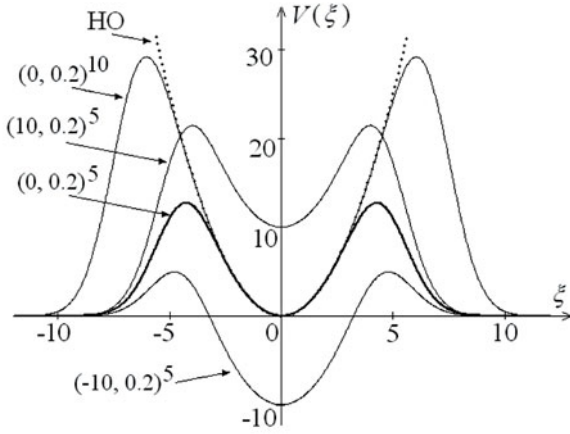


Fig. 1. Illustration of pseudo-Gaussian potentials belonging to different $(\lambda, \mu)^r$ models.

by a change of variable from x to $\xi = x/a$:

$$a = \sqrt{\hbar/m\omega}. \quad (3)$$

In equation (2), $\nu = 2E/\hbar\omega$, $\lambda = 2V_0/\hbar\omega$ and the energy is measured in units of $\varepsilon = \hbar\omega/2$. If we define the operator number by:

$$N_\lambda = -\frac{d^2}{d\xi^2} + \xi^2 + \lambda \quad (4)$$

the Schrödinger equation (2) may be written:

$$N_\lambda \psi = \nu \psi. \quad (5)$$

A PGO model [6] is a generalization of the HO model, where the PGO operator number is defined by:

$$N_{\lambda,\mu}^r = -\frac{d^2}{d\xi^2} + W_{\lambda,\mu}^r(\xi) \quad (6)$$

being governed by the dimensionless potentials:

$$W_{\lambda,\mu}^r(\xi) = \left(\lambda + \sum_{k=1}^r C_k \xi^{2k} \right) e^{-\mu \xi^2} \quad (7)$$

with $C_k = (\lambda\mu + k) \mu^{k-1}/k!$. The values of the parameters $\lambda \in \mathbb{R}$, $\mu > 0$ and $r = 1, 2, \dots$ determine the properties of PGO, reflected in the shape of the potentials as displayed in Figure 1.

$(\lambda, \mu)^r$ refers to a PGO model.

3 Computational approach

For a sequence of potential steps as in Figure 2, TMM manages to properly connect the input *in* and output *out* plane waves to the potential region:

$$\begin{aligned} \psi_{in}(\xi) &= A_{in} e^{ik_{in}\xi} + B_{in} e^{-ik_{in}\xi}, \\ \psi_{out}(\xi) &= A_{out} e^{ik_{out}\xi} + B_{out} e^{-ik_{out}\xi} \end{aligned} \quad (8)$$

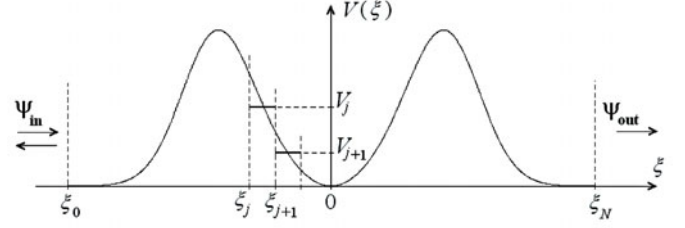


Fig. 2. Illustration to TMM. Piecewise constant potentials V_j , $j = 0 \dots N-1$, approximate the PGO model $(0, 0.2)^5$.

with BenDaniel and Duke boundary condition [10] at every interface:

$$\psi_{j-1} = \psi_j; \quad \frac{1}{m_{j-1}} \frac{d\psi_{j-1}}{d\xi} = \frac{1}{m_j} \frac{d\psi_j}{d\xi} \quad (9)$$

which guarantees current conservation. m_j is the particle relative mass in the layer j and $k_j = \sqrt{m_j(E - V_j)}$ is dimensionless wave vector.

The main quantity derived with TMM is the transmission coefficient:

$$T = J_{out}/J_{in} \quad (10)$$

as a function of energy, when the particle evolves in the piecewise constant potential. $J \propto \pm |C|^2$ is the dimensionless probability current density for a free particle described by a plane wave $C e^{\pm ik\xi}$.

For a sequence of N potential steps we can compute the transmission coefficient:

$$T = \frac{k_{out}}{k_{in}} \frac{m_{in}}{m_{out}} \frac{1}{|\mathbf{M}_{11}|^2}, \quad (11)$$

$\mathbf{M}_{11}$ is the (1, 1) element of the transfer matrix $\mathbf{M}$, which connects the *in* and *out* wave vectors:

$$\begin{pmatrix} \psi_{in} \\ \psi_{out} \end{pmatrix} = \mathbf{M} \begin{pmatrix} \psi_{out} \\ \psi_{in} \end{pmatrix} \Leftrightarrow \begin{pmatrix} A_{in} e^{ik_{in}\xi} \\ B_{in} e^{-ik_{in}\xi} \end{pmatrix} = \mathbf{M} \begin{pmatrix} A_{out} e^{ik_{out}\xi} \\ B_{out} e^{-ik_{out}\xi} \end{pmatrix}. \quad (12)$$

Since any arbitrary potential can be approximated with piecewise constant potentials the above method is general. For a complete derivation of TMM see [11].

4 Results and discussion

Theoretical study of a PGO structure reveals equidistant bond states and resonances, but these calculation are done on the hypothesis that the particle mass is equal to the one of the free electron which is not the case in physical quantum devices. A semiconductor heterostructure that exhibit a PGO build-in potential may be fabricated using epitaxial techniques, like MBE, by varying the composition x of a ternary alloy $A_x B_{1-x} C$. A survey of the semiconductor alloys properties (for example [12]) demonstrates that generally the electron effective mass m^* varies linearly or parabolic with the alloy composition. We try to answer in this paper to the following two questions: (1) what is the pattern of the build-in potential leading to

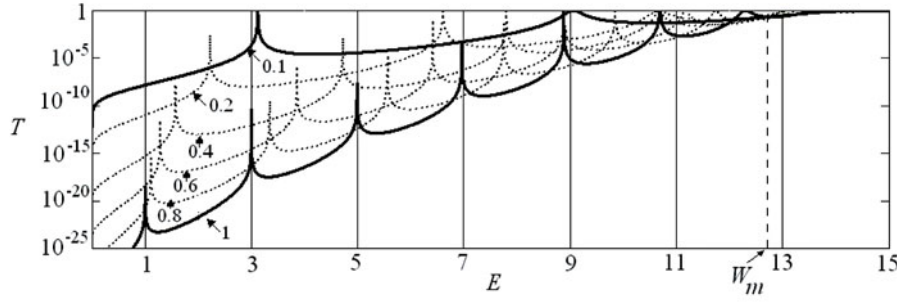


Fig. 3. Transmittance of the PGO $(0, 0.2)^5$ model as function of energy (in units of $\hbar\omega/2$). The curves parameter is the relative mass m^*/m_0 , being indicated on the graph. W_m indicate the PGO potential maximum.

the replication of the PGO electronic states by epitaxial techniques? (2) What is the procedure to find it?

To find answers, first we will show the shifting of the theoretical electron energy states system calculated in the hypothesis of constant electron effective mass taken as equal to the mean over its variation interval, when x takes values in a given domain. Second we will explain the approach for tuning the theoretical PGO potential in order to reproduce the original electron energy states system. Third, we will consider the more realistic case, when the effective mass is point dependent but considering the PGO potential calculated at the second step. A new shift in the levels position occurs but less than in the previous case, which eventually requires a new refining of the potential. Finally, aiming for replicating the PGO theoretical energy states, the method to achieve the dependence of the alloy composition x in respect to the heterostructure growth coordinate is detailed.

In the illustrations below we will refer to the ternary alloy $\text{Cd}_{1-x}\text{Mn}_x\text{Te}$ with the following properties [13]: Bandgap: $E_g = (1.606 + 1.587x)$ eV; Band alignment: 70% of total discontinuity in the conduction band, $\Delta E_{CB} = 0.7$; electron effective mass: $m^* = (0.11 + 0.067x)m_0$.

4.1 Effective mass effect on the theoretical PGO energy states system

By means of TMM we calculated the transmittances for PGO $(0, 0.2)^5$ model assuming different values for the electron relative mass. The potential has been discretized into $N = 200$ slides of $\delta\xi = 0.1$ thickness. In Figure 3, the results are compared to the theoretical transmittance ($m_r = 1$). A non-linear shift in the levels position can be observed; at $m_r = 0.1$, (close to the mean value of the ordinary ternary alloy) only two levels still exist inside. This behavior can be easily explained if we remember the equation for energy states inside the infinite quantum well: $E_n = \hbar^2\pi^2n^2/(2m^*l_w^2)$, where l_w is the well width. As m^* decreases, the separation between levels increases. The solution to preserve the levels position is to properly increase the well width.

4.2 Tuning the PGO model in the hypothesis of constant effective mass

In order to compensate for replacing the free electron mass with the average effective mass in the PGO model, the

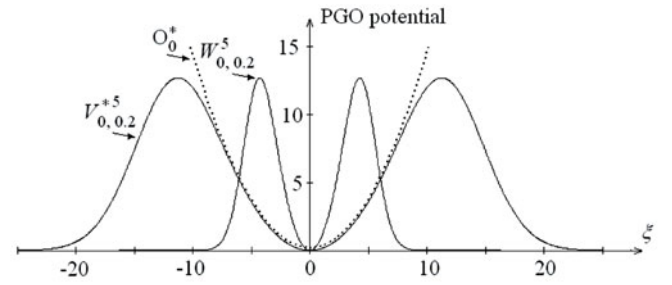


Fig. 4. The original PGO potential $W_{0,0.2}^5(\xi)$ and the HO potential O_0^* superimposed by the corrected PGO potential $V_{0,0.2}^{*5}(\xi)$, in the constant effective mass approximation $m_r = 0.1435$.

necessary changing of the theoretical PGO potential basically outcomes from the above examination of the infinite quantum well. A simple rescaling of the PGO model suffices. As we have seen in the PGO models definition (see Sect. 2), the length is measured in units of a , which is related to the particle mass by equation (3). If an effective mass is used in equation (3) the following identity occurs:

$$a^* = \sqrt{\frac{\hbar}{m\omega}} = \sqrt{\frac{\hbar}{m_0 m_r \omega}} = \frac{1}{\sqrt{m_r}} a. \quad (13)$$

Thus, the theoretical potential PGO can be simply adjusted by using the effective mass reference length a^* in equation (7):

$$V_{\lambda,\mu}^{*r}(\xi) = \left(\lambda + \sum_{k=1}^r C_k (\xi \sqrt{m^*})^{2k} \right) e^{-\mu (\xi \sqrt{m^*})^2}. \quad (14)$$

As illustration, Figure 4 displays the HO potential O_0^* , the theoretical model PGO potential $W_{0,0.2}^5(\xi)$ and the corrected one $V_{0,0.2}^{*5}(\xi)$, using the $\text{Mn}_x\text{Cd}_{1-x}\text{Te}$ alloy parameters in calculations. Because of expanding the PGO well length while preserving the slide extent to $\delta\xi = 0.1$, the number of slides N has been increased to 600. A visual inspection of Figure 4 demonstrates that the corrected potential overlay the HO potential, which guarantees the initial electronic states system.

In Figure 5, the transmittances of the original and corrected PGO potentials are plotted. It can be seen that the transmittance calculated assuming the adjusted potential

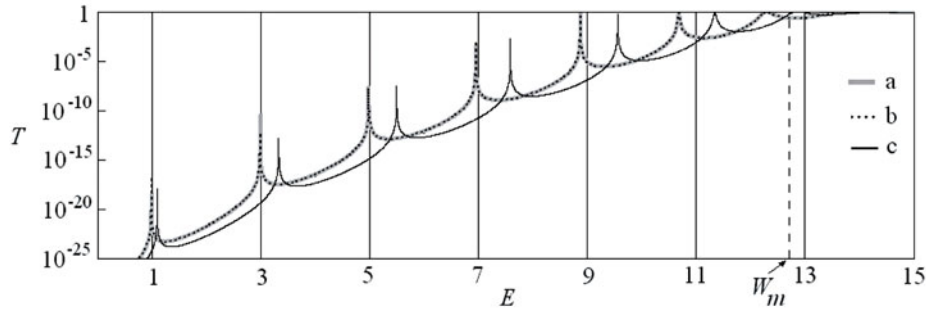


Fig. 5. Transmittance calculated for different PGO potentials: (a) $W_{0,0.2}^5(\xi)$, $m_r = 1$; (b) $V_{0,0.2}^{*5}(\xi)$, $m_r = 0.1435$; (c) $V_{0,0.2}^{*5}(\xi)$, $m_r = 0.11 + 0.067V_{0,0.2}^{*5}(\xi)/12.75$. Energy is in units of $\hbar\omega/2$.

$V_{0,0.2}^{*5}(\xi)$ exactly overlaps the theoretical transmittance. However it is just the first step in our calculations. In fact we have to solve a point depending mass problem. Assuming $V_{0,0.2}^{*5}(\xi)$ and a point dependent electron effective mass, the transmittance slightly deviates from the theoretical target, which is also shown in Figure 5. This time, the perturbation of the PGO energy states structure is smaller than in the first case when the mean effective mass was considered instead m_0 . This can be easily explained by the severe drop of the electron mass from m_0 to mean effective mass of $0.1435m_0$ in the first step while here the variation of Δm^* around mean effective mass is of $0.035m_0$. At the final step, a way to minimize the perturbation of the energy states due to point dependence of effective mass has to be derived.

4.3 Tuning the PGO model in the assumption of point dependent effective mass

From Figure 5 we may observe that taking into account point dependence of the effective mass nonlinearly shifts the levels to higher energy. Thus, the potential well should be expanded as the energy increases. To do this we work on the following procedure.

First we introduce a new correction $c(\xi)$ in the $V_{\lambda,\mu}^{*r}(\xi)$ formula:

$$U_{\lambda,\mu}^{*r}(\xi) = \left(\lambda + \sum_{k=1}^r C_k \left(\xi c(\xi) \sqrt{m^*} \right)^{2k} \right) e^{-\mu(\xi c(\xi) \sqrt{m^*})^2}. \quad (15)$$

The task consists in finding $c(\xi)$ so that the system levels of $U_{\lambda,\mu}^{*r}(\xi)$ are overlapping the original PGO ones. We discuss the numerical procedure continuing the illustration with $V_{0,0.2}^{*5}(\xi)$.

We choose to divide the PGO potential into six intervals of equal length 2. Every interval is centered on the expected values of the HO energy levels, i.e. 1, 3, 5, 7, 9 and 11. On every interval we search the suitable correction $c(\xi) = \text{const}$. In this way we simplify the general problem of finding $c(\xi)$ to one at which $c(\xi)$ exhibits a piecewise constant behavior. The coordinates ξ_i of every interface are simply determined by solving the equation $V_{0,0.2}^{*5}(\xi_i) = i$, $i = 2, 4, 6, 8$ and 10 . In every interval $c(\xi)$

has been evaluated as the transmittance peaks calculated with equation (10) lay over the corresponding HO energy state. Summing up the results from the numerical procedure optimization, the potential $U_{0,0.2}^{*5}(\xi)$ can be written:

$$U_{0,0.2}^{*5}(\xi) = \left(\sum_{k=1}^5 \frac{0.2^{k-1}}{(k-1)!} \left(\xi c(\xi) \sqrt{0.1435} \right)^{2k} \right) e^{-0.0287 \xi^2 c(\xi)^2} \quad (16)$$

where

$$c(\xi) = \begin{cases} 0.90 & \text{if } |\xi| < 3.73 \\ 0.89 & \text{if } -5.28 < \xi \leq -3.73 \quad \vee \quad 3.73 < \xi \leq 5.28 \\ 0.91 & \text{if } -6.49 < \xi \leq -5.28 \quad \vee \quad 5.28 < \xi \leq 6.49 \\ 0.92 & \text{if } -7.56 < \xi \leq -6.49 \quad \vee \quad 6.49 < \xi \leq 7.56 \\ 0.94 & \text{if } -8.65 < \xi \leq -7.56 \quad \vee \quad 7.56 < \xi \leq 8.65 \\ 1.05 & \text{otherwise.} \end{cases} \quad (17)$$

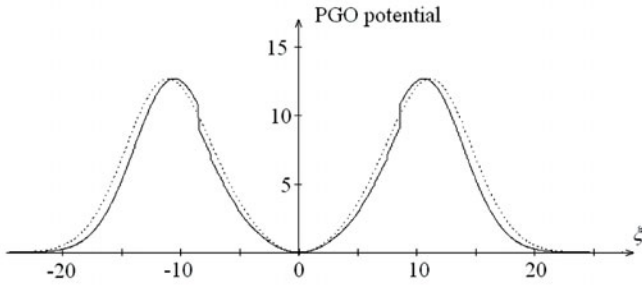
Equation (17) shows that $c(\xi)$ exhibits a non-linear behavior which means that in every interval a fitting PGO model can be set up but not for the entire domain. Equation (16) with $c(\xi)$ given by (17) is plotted in Figure 6. Unlike the sizeable difference between $W_{0,0.2}^5(\xi)$ and $V_{0,0.2}^{*5}(\xi)$ in Figure 4, the comparison of $U_{0,0.2}^{*5}(\xi)$ to $V_{0,0.2}^{*5}(\xi)$ in Figure 6 shows just a slight difference. It justifies the tacit assumption in equation (17) that in the well outside, the potential is adjusted with 1.05 (the value corresponding to its peak) leading to an irrelevant alteration of the barriers width.

The transmittance calculated with equation (10) employing the potential from equations (16–17) is plotted in Figure 7. It confirms that the aim was achieved, the electronic levels system being identical with the particular HO one.

The method flexibility for tailoring a pre-defined energy levels system in a layered structure is emphasized in Figure 8 where the transmittances calculated with equation (10) for different $c(\xi)$ are plotted in the E 4.5... 5.5 window. A visual inspection shows a shift to lower energy of the transmittance peak, from 5.5 to 4.7, as $c(\xi)$ decreases from 1 to 0.85. It is a very important observation, because the above procedure allows, by slightly altering a

Table 1. The correction $c(\xi)$ in equation (15), calculated in order to position the levels 2, 3, 4 and 5 at energy E inside a PGO $(0, 0.2)^5$ well, made from different alloys.

Material		$\text{Cd}_{1-x}\text{Mn}_x\text{Te}$	$\text{Ga}_x\text{In}_{1-x}\text{P}$	$\text{Al}_x\text{Ga}_{1-x}\text{As}$	$\text{InAs}_{1-x}\text{Sb}_x$
Mean effective mass		$0.1435m_0$	$0.096m_0$	$0.081m_0$	$0.0185m_0$
Level number	E	$c(\xi)$	$c(\xi)$	$c(\xi)$	$c(\xi)$
2	2.5	0.76	0.78	0.78	0.88
	2.75	0.83	0.86	0.86	0.96
	3	0.89	0.93	0.94	1.04
	3.25	0.98	1.01	1.02	1.12
	3.5	1.05	1.08	1.10	1.20
3	4.5	0.82	0.84	0.85	0.93
	4.75	0.87	0.89	0.90	0.98
	5	0.91	0.93	0.95	1.03
	5.25	0.96	0.98	1.00	1.08
	5.5	1.01	1.03	1.05	1.13
4	6.5	0.85	0.87	0.9	0.95
	6.75	0.90	0.91	0.94	0.98
	7	0.92	0.94	0.97	1.02
	7.25	0.96	0.98	1.01	1.06
	7.5	1.00	1.01	1.05	1.09
5	8.5	0.88	0.90	0.93	0.96
	8.75	0.92	0.93	0.97	0.98
	9	0.94	0.96	1.00	1.01
	9.25	0.97	0.99	1.03	1.04
	9.5	1.00	1.01	1.06	1.06

**Fig. 6.** The potential $U_{0,0.2}^{*5}(\xi)$ (Eqs. (16–17)) (line) superimposed over the corrected PGO potential $V_{0,0.2}^{*5}(\xi)$ (Eq. (14)) (dots).

PGO model, to precisely control the energy levels structure. In other words, by preserving a potential shape close to the pseudo-Gaussian one, the method offers the possibility to shape the energy levels system as demanded by the specific application.

4.4 Semiconductor layers system

Let us to estimate the characteristics of our device assuming that the PGO potential minimum is localized in the CdTe layer ($x = 0$) while its maximum $W_m = 12.75$ is reached in the CdMn layer ($x = 1$). Because of the linear correlation between the shift in conduction band of the $\text{Cd}_{1-x}\text{Mn}_x\text{Te}$ alloy and its composition $\Delta E_{CB} = 0.7 \times 1.587 \times x$ (in eV), equation (16) establishes a straightforward relation between alloy composition and the coor-

dinate on the growth axis. It is only necessary to convert the dimensionless equation (16) into a physical one.

Under our assumptions, the unit for energy is $\hbar\omega/2 = 0.0871$ eV and then the separation between levels is $\delta E = \hbar\omega = 0.174$ eV. Thus, our device exhibits six equally spaced levels in the well, tuned in the infrared spectrum at a wavelength $\lambda = 7.128$ μm . The characteristic length is $a^* = a/\sqrt{m_r^*} = 1.747$ nm and the total length of the device is $L = 40a^* = 69.9$ nm. We stress that the specific wavelength at which the device is working can be set a priori by a proper choice of the energy unit in the PGO model.

4.5 On the tuning algorithm limitations

In all above calculations, $\text{Mn}_x\text{Cd}_{1-x}\text{Te}$ alloy parameters have been used. The next question is, whether $c(\xi)$ as given by equation (17) is working with other materials? So we repeated the tuning procedure for $(0, 0.2)^5$ PGO model but considering other three materials: $\text{Ga}_x\text{In}_{1-x}\text{P}$, $\text{Al}_x\text{Ga}_{1-x}\text{As}$ and $\text{InAs}_{1-x}\text{Sb}_x$. The results are summarized in Table 1. Each value of $c(\xi)$ is applied to a ξ domain resulted from the equation $V_{0,0.2}^{*5}(\xi_i) = \nu \pm 0.5$. Regarding colons in Table 1 it can be seen that for a given material, as the level needs to be localized at a higher energy, an increase of the $c(\xi)$ amplitude is required; $c(\xi)$ is almost linearly related to the energy E . Looking at rows in Table 1, it can be noted that as the effective mass decreases as $c(\xi)$ shifts to higher values; however a sensible rule cannot be found. This behavior is due to the particular dependence of effective mass on the alloy composition. Consequently, the tuning procedure has to be run for each alloy.

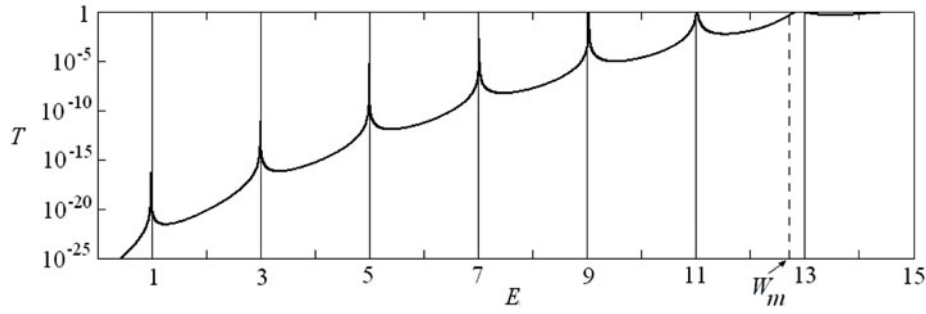


Fig. 7. Transmittance calculated with equation (11) using the potential given by equations (16–17). Energy is in units of $\hbar\omega/2$.

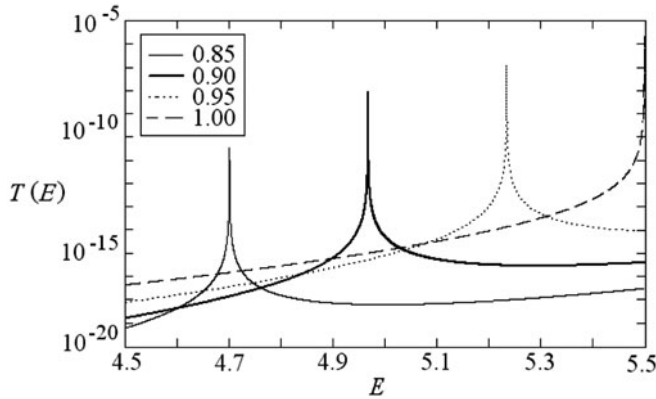


Fig. 8. Transmittance calculated with equation (11) when the potential is given by equation (16). The curve parameter is $c(\xi)$, constant.

Theoretically, there is no limitation in matching the energy levels with data inside the PGO well. However, as $c(\xi)$ drifts away from unity, so does the potential shape from the PGO one. On the other hand, when a PGO model is employed in a semiconductor structure design project, the limitations arise from the material properties only. The maximum available shift in conduction band $\Delta E_{CB, m}$ determines the maximum of the quanta. The device dimension needed to reproduce a PGO model has to be long enough to accommodate the potential gradient but also short enough to preserve the quantum description of the well.

5 Conclusions

In this paper we deal with pseudo-Gaussian models in the envelope function and effective mass approximations whose potentials combine the desirable qualities of short-range HO dynamics and Gaussian asymptotic behavior. The discrete energy levels tend to remain equidistant, largely undisturbed by the parameter values. Based on the transfer matrix method, an algorithm for adjusting the theoretical PGO potential in the more realistic hypothesis of point depending mass is put forward. The practicability of the numerical procedure is illustrated *en detail* by designing a layered structure from the ternary

alloy $\text{Mn}_x\text{Cd}_{1-x}\text{Te}$ in order to replicate the electronic states system of the pseudo-Gaussian model $(0, 0.2)^5$. It is worthwhile to mention that the above described method is not restricted to the replication of the theoretical PGO features; it is in fact general. No matter how the requested electronic states may look like, it is possible that the system may be easily tailored in a ternary alloy by applying the algorithm in a system of piecewise pseudo-Gaussian model. We expect that these virtues shared by the above described procedure make it a suitable tool for designing new quantum electronic devices in semiconductor heterostructures.

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